

Enterprise Risk Introduced by AI-Based Regulatory Authoring



AI-Driven Regulatory Authoring: Challenges, Solution, & Risk Landscape

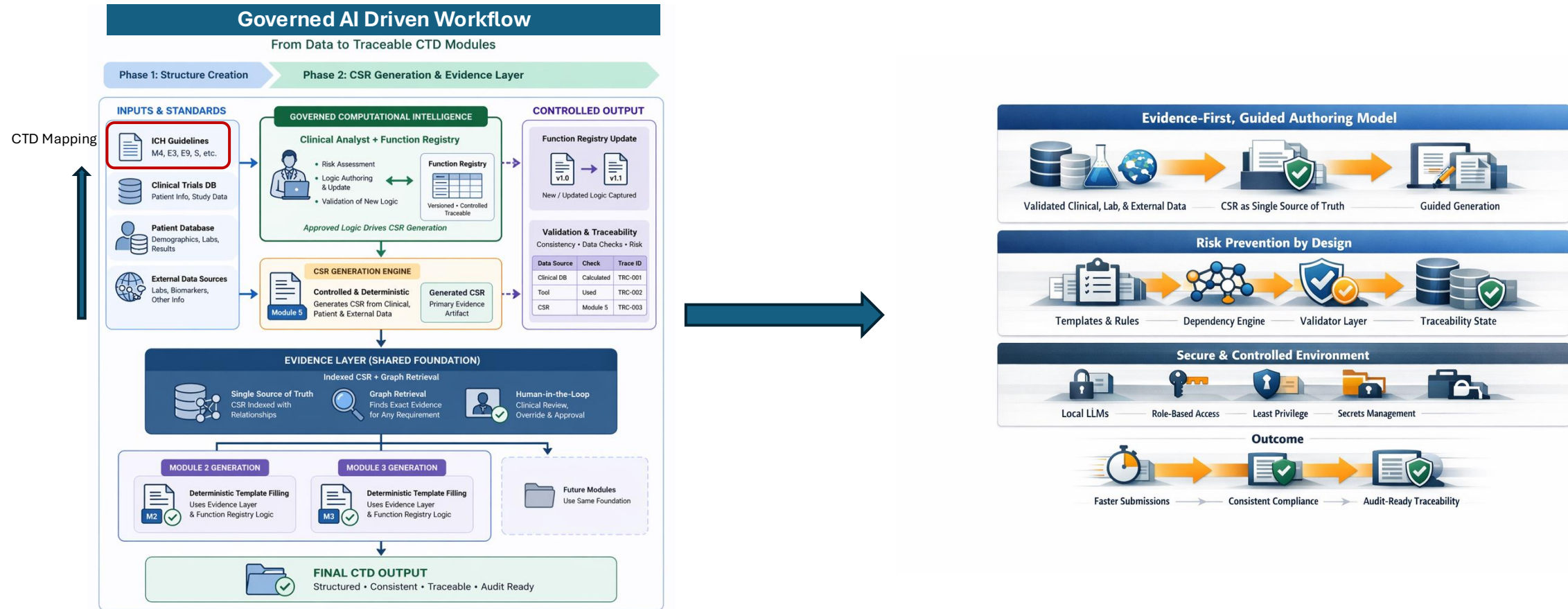


AI-Driven Regulatory Authoring

Objective

Design an AI-powered regulatory authoring platform that accelerates submissions while preserving regulatory control, consistency, and audit-readiness across the product lifecycle.

Traceability state persistence

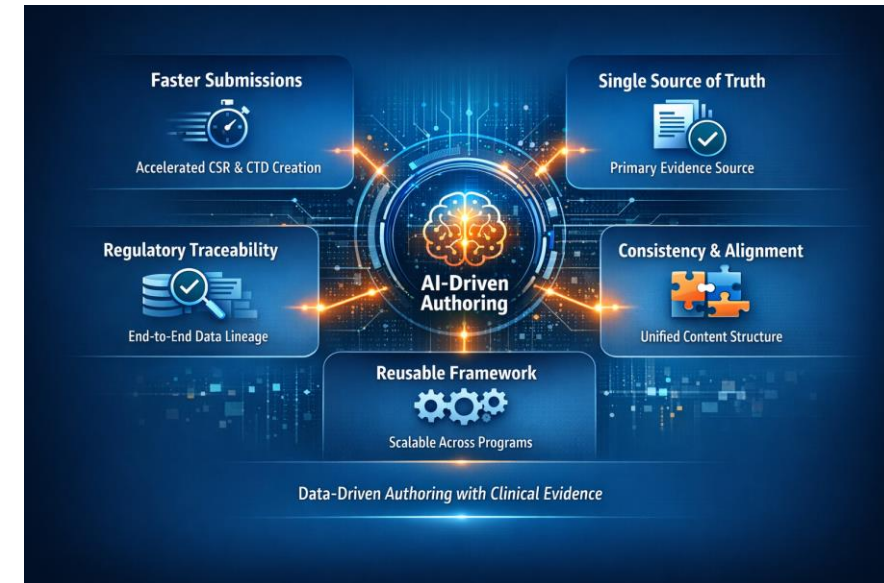


AI-driven regulatory authoring that reduces risk, not control.

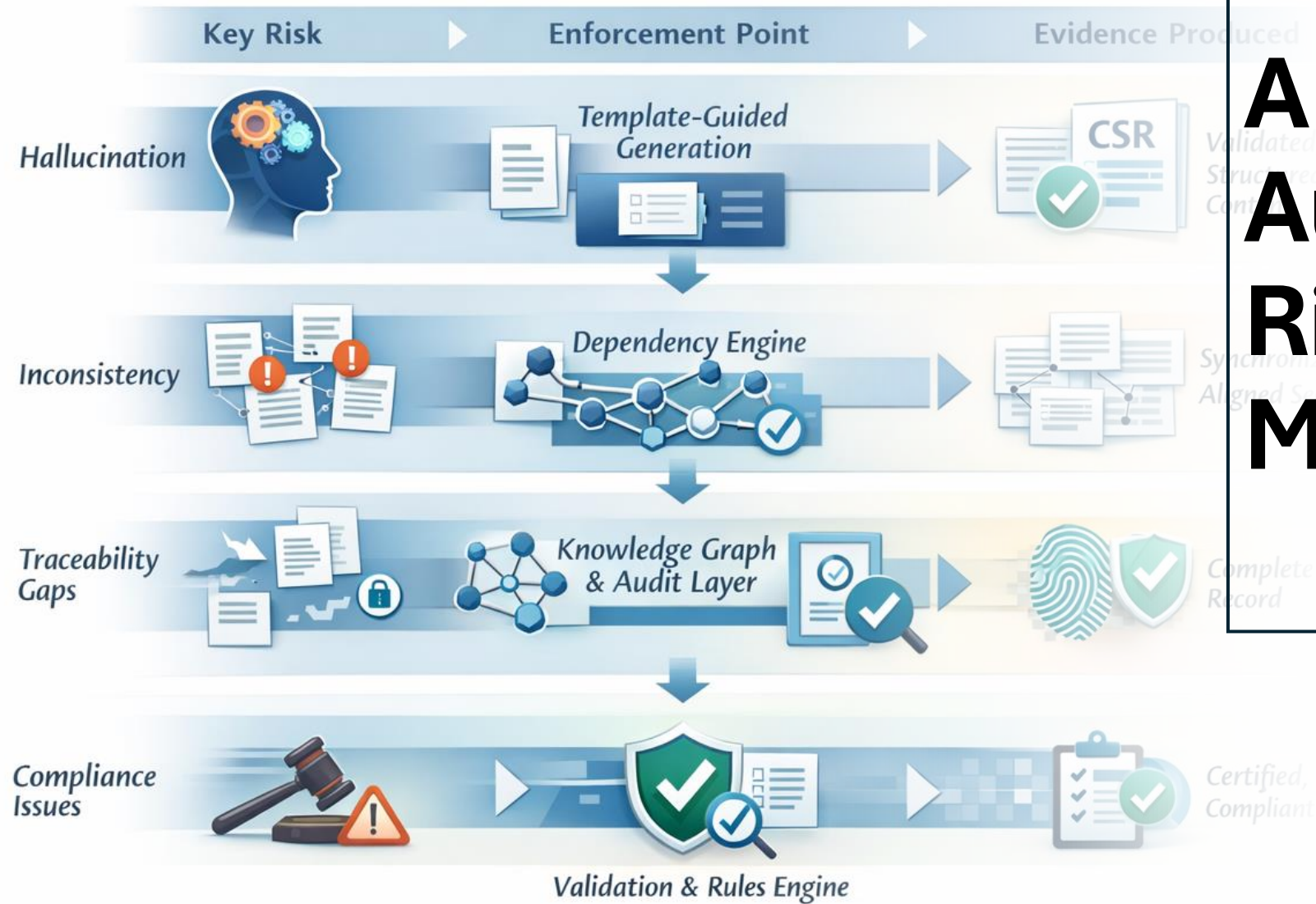
Data used in demo: Clinical Data: Bells Palsy
<https://www.kaggle.com/dillonmyrick/bells-palsy-clinical-trial>

AI-Authoring - How

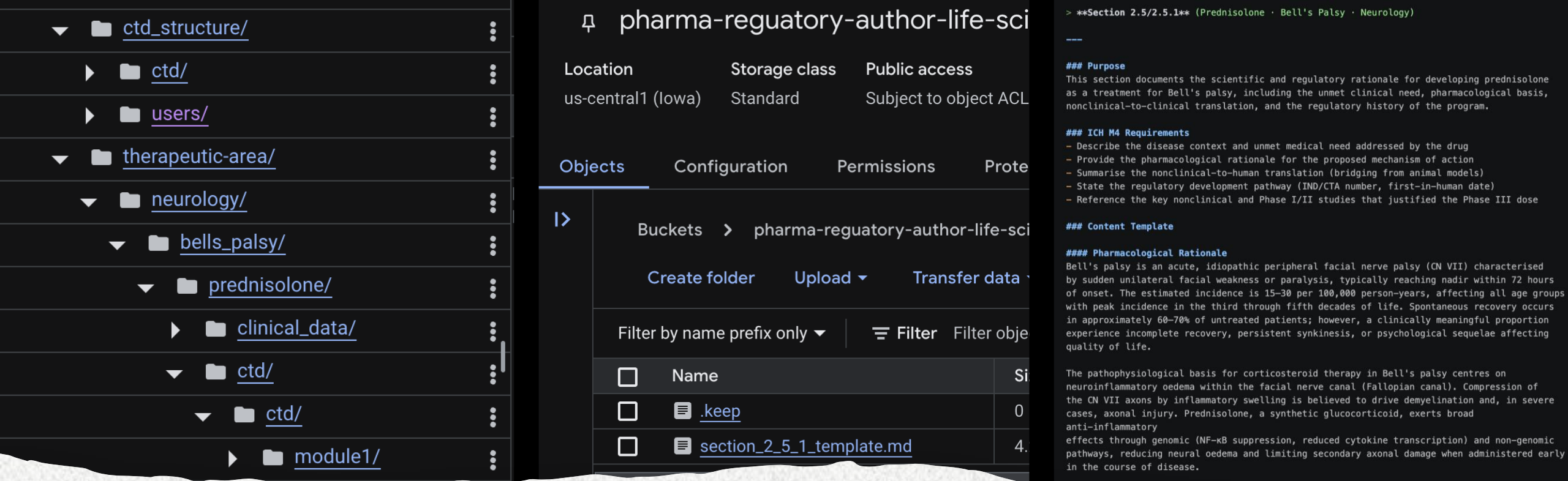
This platform does not replace regulatory judgment — it operationalizes it.



How Risks Are Mitigated by Design



**AI-
Authoring –
Risk
Mitigation**



Example: How risk is minimised

- Scalable
- Privacy protected: Local hosted LLM
- Role based Control: Access based on need to know
- Least privilege Authentication: No human error
- Secret Manager: No accidental leak of data

Traditional Authoring	AI-Driven Authoring
Manual drafting across documents	AI-assisted, guided content generation
Repetitive copy-paste	CSR as reusable evidence source
High inconsistency risk	Built-in data & semantic consistency
Limited traceability	End-to-end traceability by design
High review burden	Faster review with aligned content

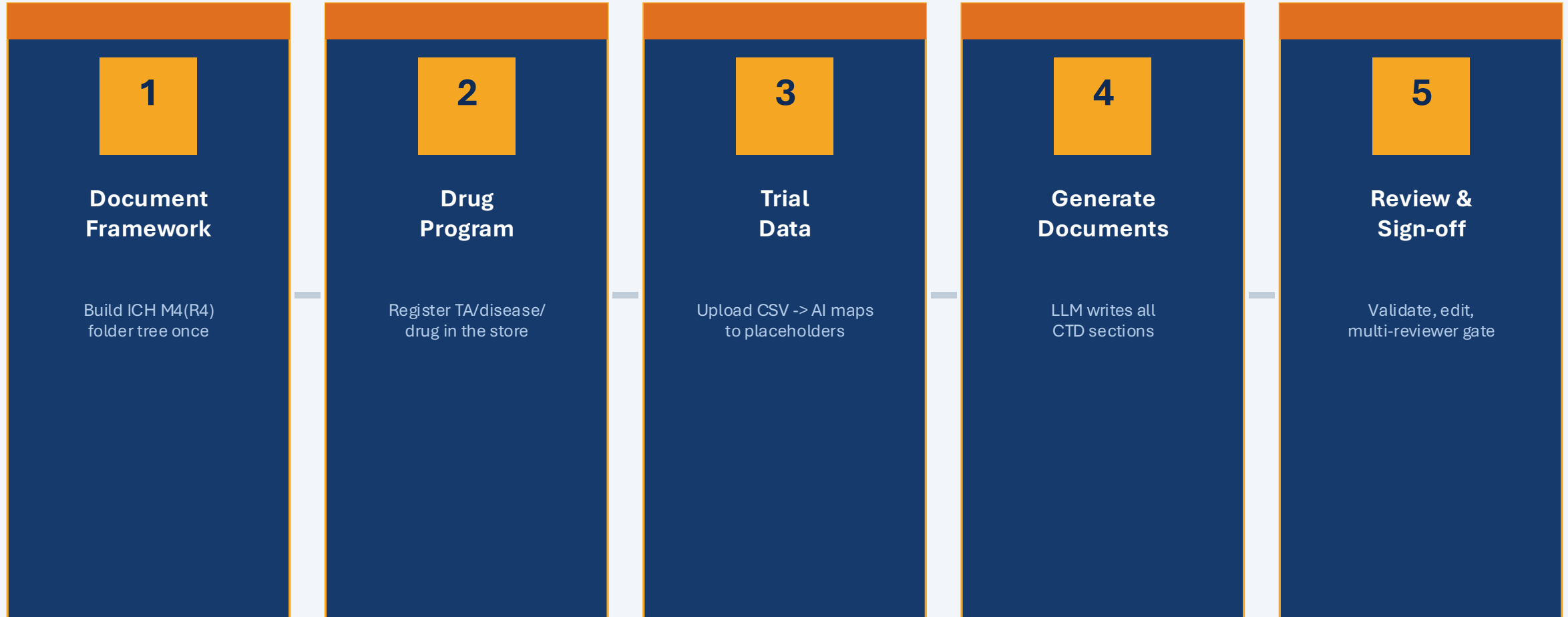
Outcome

- Faster submissions
- Less rework
- Higher quality

Risk	Your Control
Hallucination	Template + Analytical Engine
Inconsistency	Dependency Engine
Traceability	Knowledge Graph/Persistence state/validation certificate
Compliance	Validator Layer

5-Step Authoring Workflow — ICH M4(R4) Submission

Structure → Program → Data → Authoring → Review



STEP 1

Document Framework

Build & publish the ICH M4(R4) folder hierarchy

✓ Document framework published — 198 folder paths live

- ▶ Queries ICH M4(R4) guidelines to scaffold the full CTD module tree
- ▶ Creates 198 folder paths in GCS document store
- ▶ One-time setup — shared across all drug programs
- ▶ Completes in 3–8 minutes; idempotent (safe to re-run)
- ▶ Publish button makes the tree visible to downstream writers

Regulatory Authoring Platform

AI-assisted authoring for ICH M4(R4) Common Technical Documents — follow the steps below to generate your regulatory submission content.

1 · Document Framework

2 · Drug Program

3 · Trial Data

4 · Generate

5 · Review

Reviewer Identity (demo persona)

1 · Document Framework

2 · Drug Program

3 · Trial Data

4 · Generate

5 · Review

Step 1 — Document Framework

Build and publish the ICH M4(R4) folder hierarchy. Done once; shared across all programs.

✓ Document framework published

The ICH M4(R4) folder hierarchy is live and shared across all drug programs.

i This step creates the official ICH M4(R4) Common Technical Document (CTD) folder structure by querying the ICH guidelines. It takes 3–8 minutes and only needs to be done once — the result is shared across all drug programs.

Build Document Framework

✓ Publish Framework

One Time Activity

- Build Document Framework
- Publish Framework

STEP 2

Drug Program

Register the therapeutic area, indication and drug

✓ Active: Neurology / Bells Palsy / Prednisolone

- ▶ Creates a dedicated GCS namespace for this drug program
- ▶ Stores therapeutic area, disease type and drug name in session
- ▶ Pre-fills all downstream forms — no re-entry required
- ▶ Supports multiple concurrent programs per deployment
- ▶ Sanitises input to prevent API encoding issues

Regulatory Authoring Platform

AI-assisted authoring for ICH M4(R4) Common Technical Documents — follow the steps below to generate your regulatory submission content.

1 · Document Framework 2 · Drug Program 3 · Trial Data 4 · Generate 5 · Review

 Reviewer Identity (demo persona) ◀

1 · Document Framework **2 · Drug Program** 3 · Trial Data 4 · Generate 5 · Review

Step 2 — Drug Program

Create a dedicated folder for all regulatory documents for this drug and indication.

✓ Active: Neurology / Bell's Palsy / Prednisolone

Program folder is set up in the document store.

 Provide the details of the drug program you are authoring for. This creates a structured GCS folder that holds all generated CTD documents, clinical data, and status files for this specific drug and indication.

Therapeutic Area

Neurology

Disease / Indication

Bels_Palsy

Drug Name

Prednisolone

 Set Up Program Folder

- Set up project structure at program level

STEP 3

Clinical Trial Data

Upload trial CSV — AI maps columns to regulatory placeholders

(pending) Upload a CSV to enrich generated content

- ▶ Accepts any structured CSV of clinical trial results
- ▶ Clinical Analyst service auto-maps columns to ICH placeholders
- ▶ e.g. recovery_rate → {{RECOVERY_RATE_COMPLETE}}
- ▶ Mapping stored in program GCS folder for traceability
- ▶ Optional but strongly recommended for accurate content

Regulatory Authoring Platform

AI-assisted authoring for ICH M4(R4) Common Technical Documents — follow the steps below to generate your regulatory submission content.

1 · Document Framework

2 · Drug Program

3 · Trial Data

4 · Generate

5 · Review

Reviewer Identity (demo persona)

1 · Document Framework

2 · Drug Program

3 · Trial Data

4 · Generate

5 · Review

Step 3 — Clinical Trial Data

Upload your clinical trial results CSV. The AI maps columns to regulatory placeholders.

Complete Step 2 first

Upload a CSV file containing your clinical trial data. The system will automatically identify relevant columns (e.g. recovery rates, demographics, adverse events) and connect them to the correct placeholders in the regulatory document templates. This step is optional but strongly recommended for accurate content.

Clinical trial CSV file



Drop File Here

- or -

Click to Upload

- Upload CSV/ In production DB schema linkage
- Register data

STEP 4

Generate Documents

AI writes all ICH CTD sections — Module 2 + Module 5

✅ 17 sections written · Validation: 5 errors · LLM deep-check passed

- ▶ Resolves all {{placeholders}} using registered clinical data
- ▶ Generates Module 2 (Clinical Overview): sections 2.1–2.7
- ▶ Generates Module 5 (Clinical Study Reports): sections 5.1–5.4
- ▶ LLM cross-module consistency check — no further inconsistencies
- ▶ Progress visible via live poll — safe to leave and return

Regulatory Authoring Platform

AI-assisted authoring for ICH M4(R4) Common Technical Documents — follow the steps below to generate your regulatory submission content.

1 · Document Framework 2 · Drug Program 3 · Trial Data 4 · Generate 5 · Review

 Reviewer Identity (demo persona) 

1 · Document Framework 2 · Drug Program 3 · Trial Data 4 · Generate 5 · Review

Step 4 — Generate Documents

AI writes all ICH CTD sections using the trial data and ICH guidelines.

✅ Generation complete

✅ Complete — 17 sections written. Validation: 1 error(s) must be resolved before submission. LLM deep-check passed — no further inconsistencies.

 Click Generate Documents to start the AI writing pipeline. The system will: ① Resolve all placeholders using your trial data, ② Generate each ICH CTD section (Module 2 + Module 5), ③ Run cross-module consistency validation. This takes 5–15 minutes. You will see a live status update on this page.

Therapeutic Area

Neurology

Disease / Indication

Bells Palsy

Drug Name

Prednisolone

 Generate Documents

Template and Schema and Dependency based Generation

STEP 5

Review Generated Documents

Browse, validate, edit and sign off every CTD section

✓ 11 sections available for review

- ▶ Section browser: dropdown by module (Module 2 / Module 5)
- ▶ Export to Word (.docx) for offline review
- ▶ Version history with snapshot per revision
- ▶ Inline document & template editing with audit attribution
- ▶ Full 3-role sign-off gate: Statistician → Writer → QA

Regulatory Authoring Platform

AI-assisted authoring for ICH M4(R4) Common Technical Documents — follow the steps below to generate your regulatory submission content.

1 · Document Framework

2 · Drug Program

3 · Trial Data

4 · Generate

5 · Review

Reviewer Identity (demo persona)

1 · Document Framework

2 · Drug Program

3 · Trial Data

4 · Generate

5 · Review

Step 5 — Review Generated Documents

Browse every generated CTD section. Evidence sources are shown beneath each section.

[Browse your generated regulatory content](#)

i Select a section from the dropdown to read its full content. Below the document you will see **Evidence & Data Sources** — every clinical data column that was mapped to a regulatory placeholder for this program. Documents are grouped by module (Module 2 — Clinical Overview, Module 5 — Clinical Study Reports).

Therapeutic Area

Neurology

Disease / Indication

Bells Palsy

Drug Name

Prednisolone

 Load Sections

Cross-Module Validation & Consistency Report

- ✓ Drug name consistency across all sections
- ✓ Demographics & safety alignment (Module 2 vs 5)
- ✓ Unfilled {{placeholder}} detection
- ✓ ICH M4 mandatory element coverage check
- ✓ LLM deep-check: benefit-risk coherence
- ⦿ Load Report before signing — cite findings
- ⦿ 5 errors flagged — resolve before submission

placeholders, ICH M4 mandatory element coverage, and LLM deep-check for benefit-risk coherence. Click **Load Report** to fetch the latest run. Reviewers should load this report **before signing** and may cite specific findings in their approval reason.

Load Validation Report

❌ FAILED

2 error(s) must be resolved before submission. LLM deep-check passed — no further inconsistencies.

Run: 7396e0c1 · Author: system · 2026-04-27 04:23 UTC · 8 issues total

🔴 Errors 7 issues

Section	Finding	Reason
2.1_ctd_table_of_contents	3 [DATA PENDING] placeholder(s) remain — author must supply data before submission.	—
2.4_nonclinical_overview	3 [DATA PENDING] placeholder(s) remain — author must supply data before submission.	—
2.5_clinical_overview	12 [DATA PENDING] placeholder(s) remain — author must supply data before submission.	—
5.1_table_of_contents	1 [DATA PENDING] placeholder(s) remain — author must supply data before submission.	—
5.2_tabular_listing_of_all_clinical_studies	12 [DATA PENDING] placeholder(s) remain — author must supply data before submission.	—
2.1_ctd_table_of_contents	1 [DATA PENDING] placeholder(s) remain — author must supply data before submission.	—
2.5_clinical_overview	5 [DATA PENDING] placeholder(s) remain — author must supply data before submission.	—

🟡 Warnings 1 issue

Section	Finding	Reason
cross-module	Ground-truth value for 'patient_age' (44.9 ± 14.6 (n=494)) was not found in any written section — confirm figures are referenced in the appropriate CTD sections.	—

- Cross-module checks
- Mandatory coverage
- LLM validation
- Before: Silent errors
- After: Early validation & consistency

Multi-Reviewer Sign-Off — Segregation of Duties

Multi-Reviewer Validation Package · Step 5

Role 1 → Clinical Statistician

Verifies demographics, endpoints, CIs

Role 2 → Medical Writer

Verifies narrative & ICH M4 structure

Role 3 → QA / Compliance Officer

Final release gate — enforced server-side

© Each reviewer must be distinct from author

© Rejection creates immutable audit record

Use as reason

Multi-Reviewer Validation Package

A released section needs three independent approvals: **Clinical Statistician** (verifies demographics, endpoints, CIs), **Medical Writer** (verifies narrative & ICH M4 structure), and **QA/Compliance Officer** (final release gate). Each reviewer must be a different person from the author and from each other — segregation of duties is enforced server-side. Review the Consistency & Validation Report above before signing, and cite specific findings in your approval reason.

VALIDATOR-BLOCKED v? · reason: 12 [DATA PENDING] placeholder(s) remain — author must supply data before submission.

Clinical Statistician
Pending

Medical Writer
Pending

QA / Compliance Officer
Pending

I am reviewing as...
Clinical Statistician

Reason (required; min 3 chars)
e.g. Demographics match SAP v2.1; CI widths within protocol — or
missing endpoints

Sign as selected reviewer

Reject as selected reviewer

Refresh gate status

Review History (audit trail)

Regulator View (audit rehearsal)

Activity Log

- Multi-role sign-off
- Audit trail
- Version control

Audit Trail, Version History & Regulator View

Audit Trail · Version History · Regulator View · Step 5

Version History: edits snapshotted to versions/vN.md

Edit attribution: IAP identity + reason recorded

Review History: chronological approvals & rejections

Regulator View: one-click inspection bundle —

document, author, reviewer sigs, validator path,

source-data provenance, immutable prior versions

© Nothing in Regulator View is editable

Sign as selected reviewer

Reject as selected reviewer

Refresh gate status

Review History (audit trail)

Complete chronological audit trail for the selected section: every generation (re-publish), reviewer approval, and rejection — oldest first. Each event is backed by an immutable witness record in GCS.

Load Review History

12 event(s) — 2.5_clinical_overview

✓

Approval

2026-04-24 16:39:16 · run admin-se

Clinical Statistician: statistician@demo.com

Demographics match SAP v2.1; CI widths within protocol thresholds; n=494 consistent with clinical_data

✓

Approval

2026-04-25 05:23:20 · run admin-se

Medical Writer: medical.writer@demo.com

2 errors: unfilled placeholder STUDY_DURATION_WEEKS in 2.5; demographic mismatch N=496 vs N=512 in 5.3. Both acknowledged per offline QC review 2026-04-25.

↻

Generation

2026-04-25 08:32:53 · run a683e0c4

Author

Version 1 generated

↻

Generation

2026-04-25 08:38:42 · run a683e0c4

Author

Version 2 generated

↻

Generation

2026-04-25 08:58:51 · run 4150ca7e

Author: medical.writer@demo.com

Version 3 generated

↻

Generation

2026-04-25 09:02:04 · run 4150ca7e

Author: medical.writer@demo.com

Version 4 generated